

OLENYOK, Russian Federation

WMO: 241250

Lat: 68.500N

Lon: 112.433E

Elev: 722

StdP: 14.32

Time Zone: 9.00 (E09)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -60.8	(c) -56.8	(d) -63.3	(e) 0.1	(f) -60.6	(g) -59.3	(h) 0.2	(i) -57.1	(j) 14.5	(k) -7.1	(l) 12.5	(m) -13.3	(n) 2.7	(o) 110	(p) 0.827

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.3	(c) 80.5	(d) 61.5	(e) 76.0	(f) 59.5	(g) 72.0	(h) 57.9	(i) 63.8	(j) 75.6	(k) 61.5	(l) 72.5	(m) 59.4	(n) 70.0	(o) 7.0	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	77.4	67.9	56.8	70.5	66.4	54.3	64.5	64.3	29.5	75.5	27.9	73.5	26.4	69.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	14.8	13.1	-60.9	85.0	5.0	4.4	-64.5	88.2	-67.4	90.8	-70.2	93.3	-73.8	96.5
			-62.4	67.0	5.6	2.2	-66.4	68.5	-69.7	69.8	-72.9	71.0	-76.9	72.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-31.9	-27.9	-7.7	14.1	32.4	52.3	59.9	52.4	37.0	12.2	-16.3	-28.1		
	DBStd	35.24	16.37	15.49	14.64	12.36	11.23	10.47	8.60	7.38	7.58	13.55	15.48	17.27		
	HDD50	14283	2538	2181	1787	1078	555	96	15	59	394	1171	1988	2421		
	HDD65	19192	3003	2601	2252	1528	1009	399	204	396	840	1636	2438	2886		
	CDD50	633	0	0	0	0	11	164	322	132	4	0	0	0		
	CDD65	69	0	0	0	0	0	17	47	5	0	0	0	0		
	CDH74	565	0	0	0	0	0	150	375	40	0	0	0	0		
	CDH80	123	0	0	0	0	0	37	79	7	0	0	0	0		
(14) Wind	WSAvg	5.5	3.3	3.6	4.7	6.8	7.7	7.5	6.3	6.1	6.1	5.6	4.2	3.7		
(15) Precipitation	PrecAvg	11.5	0.5	0.4	0.4	0.6	0.9	1.6	2.0	1.7	1.3	1.1	0.7	0.5		
	PrecMax	16.4	1.1	0.7	1.1	1.4	2.6	4.7	5.1	5.2	2.8	1.9	1.6	1.2		
	PrecMin	6.4	0.1	0.0	0.0	0.1	0.2	0.3	0.1	0.4	0.2	0.4	0.2	0.2		
	PrecStd	2.4	0.2	0.2	0.2	0.3	0.6	0.9	1.1	0.9	0.6	0.4	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	18.5	31.4	48.3	70.3	84.7	86.2	79.7	61.0	39.8	26.1	17.0		
		MCWB	12.4	17.4	27.6	38.0	52.2	63.0	64.2	64.1	51.7	36.5	25.2	16.2		
	2%	DB	5.2	7.3	25.2	42.5	60.1	78.3	82.1	73.3	55.3	35.4	18.2	8.6		
		MCWB	4.7	6.6	22.4	34.4	46.5	59.9	62.1	58.9	47.8	33.0	17.4	7.7		
	5%	DB	-1.2	0.2	19.2	37.1	53.5	73.8	78.3	69.0	51.3	32.4	11.5	4.0		
		MCWB	-1.7	-0.5	17.1	31.6	43.2	57.2	60.7	57.4	45.1	30.7	10.7	3.4		
	10%	DB	-8.1	-5.7	14.1	32.3	48.1	68.8	74.4	64.9	47.7	29.6	5.5	-2.3		
		MCWB	-8.5	-6.1	12.3	28.2	39.8	54.9	59.4	54.8	42.6	27.9	4.9	-2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	17.7	27.9	38.7	53.3	65.0	67.4	64.9	53.5	37.2	25.5	16.2		
		MCDB	13.0	18.9	32.0	47.1	67.0	80.1	79.6	75.9	57.9	39.4	26.3	16.9		
	2%	WB	4.6	6.7	22.5	34.9	47.6	61.6	64.9	61.1	49.1	33.3	17.5	7.7		
		MCDB	5.1	7.1	24.8	41.2	59.2	75.2	76.5	69.5	53.5	34.7	18.2	8.4		
	5%	WB	-1.7	-0.4	17.1	32.0	43.7	59.2	62.8	58.6	45.6	30.9	10.8	3.5		
		MCDB	-1.3	0.3	19.1	36.9	53.2	70.8	73.4	66.4	49.8	32.4	11.5	4.1		
	10%	WB	-8.4	-6.2	12.3	28.3	40.3	56.2	60.9	56.1	43.4	27.8	4.9	-2.9		
		MCDB	-8.0	-5.8	14.0	32.2	47.2	67.0	71.8	63.5	47.3	29.5	5.5	-2.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.3	16.7	17.1	13.7	15.0	15.3	14.1	10.5	9.9	11.3	11.6		
		MCDBR	15.4	16.9	17.0	18.0	18.4	20.7	21.2	20.8	16.8	10.9	15.3	15.0		
		MCWBR	15.2	16.8	15.0	13.2	10.5	10.3	8.6	10.7	10.3	9.6	15.1	14.7		
		MCDBR	15.3	17.0	16.9	17.4	17.2	19.0	16.8	17.6	14.2	10.6	15.3	15.2		
	5% WB	MCWBR	15.1	16.7	15.1	13.4	11.0	10.0	8.7	10.4	9.6	9.7	15.0	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.186	0.224	0.282	0.311	0.336	0.355	0.414	0.401	0.304	0.234	0.176	N/A		
	taud		1.868	2.023	2.061	2.093	2.214	2.420	2.246	2.227	2.464	2.142	1.888	N/A		
	Ebn at Noon		22	177	232	261	269	268	242	229	234	172	20	0		
	Edh at Noon		2	15	28	36	36	31	36	32	19	13	2	0		
(44) All-Sky Solar Radiation	RadAvg		8	175	684	1331	1749	1760	1571	1065	561	200	29	0		
	RadStd		1	11	39	77	122	127	115	134	64	23	3	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page